ARPAN KUMAR

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SKILLS

- **Languages :- Python(Primary), JavaScript (ES6+)**
- **AI / ML & Deep Learning :- TensorFlow, Keras, scikit-learn, Pandas, Numpy; Implementation of Deep Learning Architectures (CNN, LSTM, Transformers); Natural Language Processing (NLP) techniques including tokenization and text preprocessing, NLTK; Feature Engineering, Model Training, Validation, and Performance Optimization**
- **LLM & Generative AI Tools :- LangChain, Retrieval-Augmented Generation (RAG), Hugging Face Transformers, Hugging Face Inference API, OpenAI API, Prompt Engineering, Embedding Models (BGE, Sentence Transformers), Conversational Memory Systems, Pydantic**
- **Frontend & AI Interfaces :- Streamlit, React.js**
- **Backend & API Development :- FastAPI, Flask, Node.js, REST APIs**
- **Vector Databases & Data Systems :- ChromaDB, FAISS, MongoDB, MySQL**
- **Tools & Platforms :- Google Colab, kaggle, Postman, Virtual Environments**
- **Deployment & DevOps: Docker, Docker hub, AWS (EC2 – deployment), Git & GitHub, CI/CD Basics, Environment Management, API Deployment**

ACADEMIC PROJECTS

- **Machine Learning Project :**

  1. **Customer Churn Prediction System | GitHub**
     - Developed an end-to-end Customer Churn Prediction system using Machine Learning models including Logistic Regression, Random Forest, Gradient Boosting, and XGBoost.
     - Built a robust ML pipeline with feature engineering, SMOTE for handling class imbalance, and model evaluation using ROC-AUC metrics.
     - Deployed a production-ready application with FastAPI (REST API) and Streamlit (interactive dashboard), containerized using Docker and deployed on AWS.
  2. **Movie Recommendation System | GitHub | Live Demo**
     - Built classical movie recommendation system using User-based, Item-based similarity and Matrix Factorization (SVD).
     - Implemented scalable REST API using FastAPI and interactive UI using Streamlit.
     - Evaluated model using RMSE and deployed full-stack application with Docker.

- **Deep Learning Projects :**

  1. **Image Caption Generator | GitHub**
     - Developed an end-to-end deep learning system using a CNN-LSTM architecture (TensorFlow/Keras) for image caption generation, trained on the Flickr8k dataset.
     - Built a modular inference pipeline with FastAPI for real-time predictions and an interactive Streamlit UI for image upload and caption visualization.
     - Containerized the application using Docker to enable scalable and production-ready deployment.

2. **Transformer Based English to Hindi Language Translation System | GitHub | Live_Demo**
   - Developed a production-ready neural machine translation system using transformer-based models, fine-tuning MarianMT for English–Hindi translation on parallel corpora.
   - Built scalable REST APIs using FastAPI and an interactive Streamlit interface for real-time translation.
   - Designed a modular architecture for training and inference, and containerized the application using Docker for efficient deployment.

- **GenAI Projects :**

1. **Youtube-Chat-Bot (RAG-Based Video QA System) | GitHub**
   - Developed a RAG-based YouTube Chatbot enabling conversational querying of video content by extracting transcripts and performing semantic search using ChromaDB and BGE embeddings.
   - Built an end-to-end LangChain pipeline integrating embedding models and LLMs to generate context-aware, transcript-grounded responses.
   - Designed a modular, production-ready architecture with secure API key management and scalable vector storage, supporting easy deployment via APIs or Docker.
2. **RAG Pdf Chatbot Project | GitHub**
   - Developed an end-to-end RAG-based PDF chatbot using LangChain, FAISS, and Hugging Face LLMs for context-aware question answering.
   - Implemented semantic search with vector embeddings and conversational memory, enabling accurate and multi-turn interactions.
   - Built a production-ready system with FastAPI backend and Streamlit frontend, containerized using Docker for scalable deployment.

- **AI Agent Project :**

1. **AI-AGENT (Research And Task Automation) | GitHub**
   - Developed an AI agent using LangChain integrating Retrieval-Augmented Generation (RAG), tool usage, and conversational memory for context-aware task execution.
   - Implemented a Flask-based API and an interactive Streamlit frontend to enable real-time user interaction with the agent.
   - Containerized the application using Docker to ensure scalable and production-ready deployment**.**

EDUCATION

- PG Diploma in Advanced Computing | C-DAC ACTS, PUNE *(2024)*
- Bachelor of Technology | SRMSCET, BAREILLY *(2017)*
- XII (CBSE) | LPS, LUCKNOW, U.P *(2012)*
- X (ICSE) | ST. ANN'S CONVENT, LUCKNOW, U.P *(2010)*